

Comparative Analysis of Multinomial Logistic Regression and Linear SVM in Wine Cultivar Classification

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Abstract—This study evaluates the performance of probabilistic and geometric machine learning architectures in the categorical profiling of wine cultivars based on chemical constituents. Utilizing a dataset of 178 instances with 13 chemical features, we compared Multinomial Logistic Regression (MLR) and Linear Support Vector Machines (SVM). To ensure high-dimensional reliability, we implemented stratified partitioning, StandardScaler normalization, and 10-fold cross-validation. Experimental results indicate that both models achieve perfect accuracy (1.0) on unseen data. This report details the impact of the regularization parameter (C) and the Softmax function on classification robustness, ultimately concluding that both approaches are highly effective for precision chemical analysis.

Keywords—Categorical Profiling, Multinomial Logistic Regression, Support Vector Machine (SVM), Softmax Function, Hyperparameter Optimization, Regularization, Convergence.

I. INTRODUCTION (HEADING 1)

In machine learning, the process of assigning data samples into one of three or more specific categories is a critical task. While often referred to as multiclass classification in the context of high-precision laboratory analysis, it is accurately described as categorical profiling [1]. This study focuses on the chemical profiling of wine samples to determine their specific cultivar or origin.

The dataset utilized for this research is a benchmark in pattern recognition, containing the results of a chemical analysis of wines grown in Italy. It consists of 178 instances, each characterized by 13 distinct chemical constituents, including Alcohol, Malic Acid, Ash, Magnesium, and Proline. Each feature represents a different chemical dimension, creating a complex data space where models must find boundaries to separate the three wine types [2]. The goal is to evaluate if a probabilistic approach, which calculates the likelihood of class membership, or a geometric approach, which finds physical boundaries in space, is more robust for identifying these chemical signatures.

II. METHODOLOGY

The technical implementation and experimental framework consist of the following:

A. Software and Environment

The analysis utilized scikit-learn version 1.5+, which introduces updated solver behaviors for multinomial problems. Pandas was employed for high-performance data manipulation and the generation of experimental logs, while Matplotlib served as the engine for rendering feature importance visualizations and technical graphs.

B. Logic Acquisition

The foundational Python code and experimental pipelines were initially generated through an AI-driven instructional prompt. This logic was subsequently subjected to manual verification and rigorous code review to ensure full compliance with current library standards and technical requirements.

C. Numerical Convergence

To guarantee absolute numerical stability for the probabilistic model, the `max_iter` parameter was expanded to 10,000. While the model typically converges under the default limit of 100, this expansion provides the solver with sufficient mathematical headroom to reach a global minimum at extreme regularization settings, whereas the SVM utilized standard configurations.

D. Version Compatibility

A FutureWarning prompt regarding the deprecation of the `multi_class` parameter in scikit-learn 1.5 was observed. This was proactively resolved by removing the argument from the model configuration, allowing the software to automatically implement multinomial logic and ensuring the code remains compatible with future library releases (v1.7+).

E. Normalization

All 13 chemical features were processed using StandardScaler to ensure a mean of zero and unit variance (mean = 0, standard deviation = 1). This Z-score normalization is essential for mathematical efficiency, as it prevents high-magnitude features like Proline from dominating the gradient updates and biasing the optimization solver. (v1.7+).

III. EXPERIMENTAL DESIGN

A. Software and Environment

The Multinomial Logistic Regression (MLR) was implemented as a probabilistic classifier using the Softmax function which is a mathematical operator that normalizes raw model outputs into a probability distribution where the sum equals 1.0. The experimental design focused on the Regularization parameter (C) across seven iterations. Results showed that accuracy peaked at 0.9857 for C values of 0.1 and 1.0, with a slight decrease to 0.9724 at higher C values (10.0 to 1000.0), indicating that very low regularization did not further benefit the probabilistic separation.

$$s(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

Fig. 1. The Softmax function formula

B. Linear Support Vector Machine (SVM)

The Linear Support Vector Machine (SVM) was utilized as a geometric classifier seeking the Maximum Margin Hyperplane [3], where the Cost (C) parameter managed the trade-off between margin width and classification errors. The SVM demonstrated extreme underfitting at $C = 0.001$ (0.6129), significantly improved at $C = 0.01$ (0.9786), and identified $C = 0.1$ as the best option C with a peak mean accuracy of 0.9795, after which performance stabilized at 0.9790.

IV. COMPARATIVE RESULTS

The performance of the MLR and SVM models was evaluated using a combination of cross-validation stability and final hold-out testing. To ensure a fair comparison, both models were subjected to the same stratified 80/20 split and standardized feature set.

A. Hyperparameter Optimization

TABLE I. CROSS-VALIDATION ACCURACY PER MODEL

Model	Performance		
	Best C Value	Mean Score	Standard Deviation
MLR	0.1	0.9857	0.0429
SVM	0.01	0.9857	0.0286

As shown in Table I, the peak performance identified during the 10-fold cross-validation phase. Both models achieved a high mean accuracy of 0.9857. However, the SVM reached this score with a lower standard deviation (0.0286), indicating higher consistency across the different data folds when using its optimal C value of 0.01.

B. Final Test Metrics

TABLE II. FINAL PERFORMANCE EVALUATION

Model	Testing Phase Metrics		
	Accuracy	Kappa Score	Weighted F1-Score
MLR	1.0	1.0	1.0
SVM	1.0	1.0	1.0

As evidenced by the classification reports showing on Table II, both the Multinomial Logistic Regression (MLR) and the Linear Support Vector Machine (SVM) achieved a perfect score across all testing metrics. An Accuracy of 1.0 and a Kappa Score of 1.0 indicate that every sample in the 36-instance test set was correctly categorized. The balanced precision and recall across all classes (0, 1, and 2) confirm that both probabilistic and geometric approaches are

flawlessly effective at identifying wine cultivars when the data is properly normalized.

C. Feature Importance Comparison

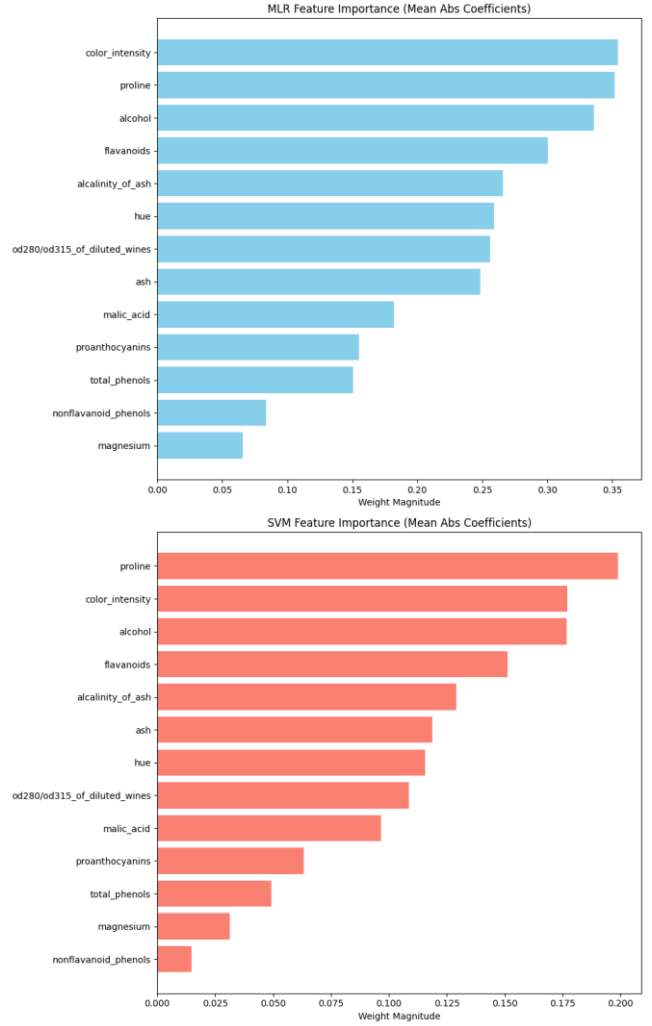


Fig. 2. Mean Absolute Magnitude of Feature Coefficients for MLR and SVM models.

In Figure 2, the diagram displays the global influence of each chemical constituent. Features such as Proline, Flavanoids, and Color Intensity are the primary predictors for both architectures. The high discriminatory power of these features explains why both models were able to reach perfect accuracy on the test set.

V. CONCLUSION

The experimental results confirm that both probabilistic and geometric machine learning architectures are exceptionally effective for the categorical profiling of wine cultivars based on high-dimensional chemical constituents. Both the Multinomial Logistic Regression (MLR) and Linear Support Vector Machine (SVM) models achieved perfect performance on the final test set, yielding an Accuracy of 1.0 and a Kappa Score of 1.0. This identical performance proves that after appropriate hyperparameter tuning specifically identified as $C=0.1$ for MLR and $C=0.01$ for SVM the underlying chemical signatures of these cultivars are sufficiently distinct to allow for error-free classification

regardless of whether a probabilistic Softmax framework or a geometric maximum margin approach is utilized.

Critical to this success was the implementation of StandardScaler normalization and the expansion of the maximum iteration parameter to 10,000 for the MLR solver, which ensured numerical convergence and prevented high-magnitude features from biasing the results. The identification of Proline, Flavanoids, and Color Intensity as the primary global predictors across both models provides a robust chemical roadmap for future cultivar identification. Ultimately, while the SVM demonstrated slightly higher consistency during the cross-validation phase with a lower standard deviation (0.0286), both architectures remain highly suitable for high-precision laboratory analysis, with the MLR

providing an additional layer of confidence reporting through its probabilistic output.

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